IT599 Project Proposal

Felicia Ainslie

Purdue University Global

IT599 Project Proposal

Remote Initiative for a VoIP Infrastructure

Introduction

The intentions of this project proposal are to update the existing security measures so that VoIP Fan's Infrastructure is secured for employees who are in the office or working remotely. The organization has determined that certain positions will soon be allowed to work from home; however, this has not been permitted in the past. As the project manager responsibilities will include creating and maintaining the project budget and schedule, researching and reviewing security recommendations, forming a communications plan, reviewing change requests, monitoring for issues, controlling the project to ensure milestones are met, and assessing the project at completion. The subsequent sections within this proposal will provide a bit of background on VoIP Fan, the issue to be addressed, how this project will address said issue, relevant stakeholders, expected expenses, anticipated milestones, and potential risks associated to the project.

Background of Organization

Voice over internet protocol provider, VoIP Fan, has over 100,000 customers; roughly 1,000 employees, and facilities located throughout the United States, Mexico, and Canada. The organization prides itself on the services it offers to a variety of organizations which include government agencies, hospitals, and restaurants among other industry types. Clientele rely on VoIP Fan for their call centers, call detail reports, call recordings, and fax services which requires the organization to ensure security, integrity, and redundancy remain top priorities. VoIP Fan was established in 2010, so it is young compared to some of its competitors; nevertheless, it strives to be the top choice for both clientele and employees.

Problem

VoIP Fan wishes to remain competitive with other providers by offering employees the opportunity to work from home. This will only be offered to the support department currently which consists of roughly 400 employees. Since this is a new initiative, there are no policies established to define the constraints or expectations for employees. Furthermore, the organization is not prepared to allow the employees to take the computers home or access company resources from their home networks. Additional issues arise as the organization is its own VoIP provider, so employees must have their home networks evaluated to determine if they are suitable candidates for remote work. This is critical as clientele rely heavily on the reliability of the support department and dropped calls or poor audio quality will damage the confidence the organization has worked so hard to build.

Purpose of Project

The intent of this project is to allow employees to assist VoIP Fan clientele from the comfort of their home. This will allow the organization to remain competitive in the market, reduce their carbon footprint, and satisfy the growing requests for remote work from existing employees. Without this project there would be no security measures to move forward with allowing remote work which could potentially cause the organization to lose valuable employees to competitors who are allowing this.

Stakeholders

Stakeholders	Responsibilities
Project manager Ms. Ainslie	• Develop and manage project budget. • Develop and manage project schedule. • Develop a communications plan. • Document any changes to project, budget, schedule, or staff members. • Hold weekly status meetings to discuss and document successes or failures. • Maintain control of the project to avoid scope creep. • Assess and document project results at conclusion.
Systems Analyst Mr. Lyst	• Analyze the existing network topology and security measures. • Research potential solutions. • Evaluate infrastructure needs for accessibility and security. • Develop documentation for new employee setup. • Assist with training remote employees.
Software Engineer Mr. Micro	• Confirm new software products are compatible with existing software. • Install new software on remote employee devices. • Confirm new software is functional.
Network Engineer Mrs. Wan	• Ensure the organization has the necessary hardware to allow staff to work remotely. • Analyze the staff home networks to ensure they are compatible with VoIP. • Document any required changes to home network configurations or hardware. • Assist with making minor configuration changes with remote staff approval.
Quality Assurance Ms. Perfect	• Develop a plan that outlines what will be tested and how it will be tested. • Have users who are unfamiliar with the system test it to determine what changes or training may be needed. • Perform penetration testing on the remote user's devices to ensure the security measures remain operational in the office and at the home network. • Analyze the reports of all tests. • Report any negative discoveries promptly. • Develop an action plan in the event of failure.

Table 1. This table describes the responsibilities of the stakeholders in the remote staff initiative.

High Level Costs Associated with Project

The following expense chart was made under the assumption that 200 of the 400 employees within the support department would be working remotely and 30% of those remote employees will require some configuration or equipment changes to their home network. Labor cost estimates were created under the expectations that employees will be paid the median rate for their position, the project is completed within 4 weeks, and final assessments are made within a week of the project's completion

Expense Type	Enterprise Cost	Total Cost
Virtual Private Network	$40/Gateway	$8,000
Mobile Device Management	$48/device	$9,600
Web Cams	$70/device	$14,000
Replacement routers	$149/device	$8,940
Extended Ethernet Cables	$31.50/Cable	$6,300
Labor	PM - $20/hr | 200 hr SA - $35/hr | 160 hr SE - $42/hr | 80 hr NE - $41/hr | 120 hr QA - $36/hr | 80 hr	PM - $4,000 SA - $5,600 SE - $3,360 NE - $4,920 QA - $ 2,880
Total Cost:	$158-307/user *Does not include labor costs.	$67,600

Table 2. This table offers a rough estimate of the total costs for the remote staff initiative (Perimeter 81, n.d.; G2.com, Inc, n.d.; Cunningham & Pinola, 2020; Cisco, n.d.; Cables.com, n.d.; ZipRecruiter, n.d.).

Expected Outcomes of the Project

All computers will have VPN and MDM software installed prior to staff members transitioning to remote work. This software will require testing to ensure end-users cannot disable it, that it remains available, and updates are automated. Logins for the VPN will also be required for each user. Documentation and training will be provided to ensure the users understand how the software operates and how common issues can be resolved. An acceptable

use policy will also be created and employee consent will be mandatory for any employee who wishes to work remotely. Remote staff will be provided web Remote staff so they may participate in team meetings or assessments virtually. These web cams must be tested to ensure the devices are properly configured and functional while not compromising the audio on each device. All home networks will be assessed to determine if changes in configuration or equipment are necessary. For some remote staff this will require on-site assistance for new routers to be installed. After the transition, quality assurance will assess the remote setup to ensure that support agents are able to access all resources needed to perform their daily operational tasks, yet unable to access resources they are unauthorized for. Additional factors that will be tested for include ease of use, bandwidth, latency, jitter, and overall equipment performance to ensure the same standards applied to in office staff members are applied to remote users.

Risks to Project Completion and Proposed Mitigations

A potential risk of this project is that remote employees may experience down time due to connectivity or hardware issues. To mitigate this risk, the organization could invest in hotspots for remote employees to use as a backup internet source and remote desktop software that allows help desk to assist staff without having to be on-site. Another risk would be that internet service providers do not support the provided replacement routers, but do not offer a router that supports VoIP. As an example, routers that cannot have SIP ALG disabled would be unsupported by VoIP Fan. This can be mitigated by contacting the ISP used by remote staff to confirm which equipment is supported prior to making purchases. One more risk would involve the lack of access to resources. This can be mitigated by working with the support department’s leadership

team to develop a comprehensive list of what staff must access to perform daily tasks as well as tasks that may not occur often, but are still required in certain circumstances.

Conclusion

The remote work initiative will offer many benefits to both VoIP Fan and staff members. This project is estimated to take four weeks and will allow 200 support agents to work from home, thus offering staff more time for their personal lives by removing the time committed to commuting to the office. The success of this project is crucial to the future of the organization as it will be a determining factor if other departments will be permitted to work remotely as well. Furthermore, this will assist the organization with remaining competitive as more organizations are offering remote positions in a variety of industries. The estimated expense of this project is just over $67,000; although, if risk mitigation is implemented, as recommended, the total project cost will rise. Still, the project will remain within the allotted budge of $250,000 which will allow excess funds to be used towards the implementation of other departments transitioning to remote positions in the future.

References

Cables.com. (n.d.). 100Ft Cat6 shielded ethernet cable snagless blue. Retrieved from https://www.cables.com/100ft-cat6-shielded-ethernet-cable-snagless-blue.html?gclid=Cj0KCQiApsiBBhCKARIsAN8o_4gkOBbqBJq96qRcxMewlAypCRMAv9UQwrhlJrlFRRaTxr-__6ml3JQaAg4CEALw_wcB

Cisco. (n.d.). Small business Routers. Retrieved from https://www.cisco.com/c/en/us/solutions/small-business/networking/routers.html#~products

Cunningham, A. & Pinola, M. (2020). The best webcams. Retrieved from https://www.nytimes.com/wirecutter/reviews/the-best-webcams/

G2.com, Inc.. (n.d.). Scalefusion. Retrieved from https://www.g2.com/products/scalefusion/pricing

Perimeter 81. (n.d.). Choose your perimeter 81 plan. Retrieved from https://www.perimeter81.com/pricing

ZipRecruiter. (n.d.). Real salaries from real employers. Retrieved from https://www.ziprecruiter.com/Salaries